

Cleaning the Signal Before Trading It

A three-pass duration-neutral backtest using WRDS enhanced end-of-month bond data

0.39%	-1.98%	1
Selected clean gross annualized return	Selected return after 25 bp turnover cost	Isolated month-end mark excluded

The conclusion changed after checking the 2025 jump bond by bond. One Apple price fell from 91.76 to 25.17 and returned to 92.24 the following month. I exclude that mark before fitting the curve and before calculating returns, then keep the untouched backtest as a sensitivity check.

Course	Instructor	Sample	Issuers
MTH 9897 - Systematic Trading	Professor Misha Boroditsky	Aug. 2022 to Mar. 2025	AAPL, AMZN, BA, CAT, DIS, T

Prepared from the executed assignment notebook and the two supplied WRDS files.

Executive Summary

My original results looked unusually strong after the start of 2025. Rather than deleting the large return itself, I traced its bond-level source. The jump comes from Apple CUSIP **037833BX7**, whose month-end price temporarily drops by 72.6% and then rises by 266.5%. Its surrounding observations are stable, and the February yield and spread also move to implausible levels relative to the adjacent months.

Answer to the data question: yes. This specific mark should be excluded before both curve fitting and return construction. It is a price-data error, not an economically credible one-month move.

Why the order matters

- 1. Curve fitting.** The February observation carries a 19.66% yield and a 466 bp Treasury spread. Leaving it in the issuer cross-section can alter the fitted Apple curve, the residual ranking, and which bonds are called rich or cheap.
- 2. Return construction.** The same mark creates a very large loss into February and a very large gain out of February. Those are two contaminated monthly returns, not one harmless outlier.
- 3. Gap treatment.** After removing February, January connects to March across 59 days. The notebook does not turn that into a monthly return because its valid next-month window is 25 to 45 days.

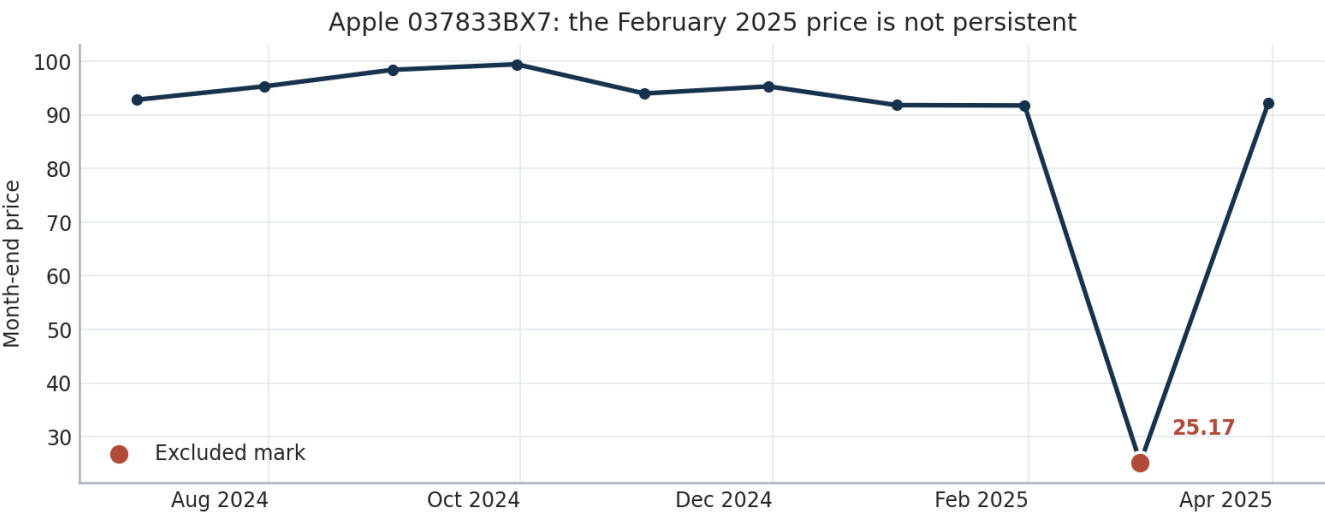
Revised result

0.15% Pass 1 clean gross	0.39% Pass 2 clean gross	0.10% Pass 3 clean gross
--	--	--

Pass 2 remains the strongest cleaned specification, but its 0.39% gross annualized return is not enough to cover the assumed trading cost. At 25 bp per unit of one-way turnover, the annualized result is -1.98%. The honest interpretation is therefore weaker but more useful: the residual signal is a sensible research framework, yet this sample does not support a profitable implementation after costs.

1. Price-Quality Audit

The cleaning rule is mechanical. Within each bond, I flag a month only when its price differs by more than 40% from both adjacent monthly observations, the two neighboring prices are within 20% of each other, and both date gaps are between 25 and 45 days. Across 7,559 otherwise eligible bond-months, it flags exactly one row.



Issuer	CUSIP	Date	Prior price	Flagged price	Next price	Yield	T-spread
AAPL	037833BX7	2025-02-28	91.76	25.17	92.24	19.66%	466 bps

Why I treat this as a bad mark

The surrounding prices are 91.76 and 92.24, only 0.52% apart, while the reported February price is 25.17. The WRDS return agrees mechanically with that price path, so the formula is not the source of the problem. The source is the underlying month-end mark. The security is Apple's 4.65% senior note due February 2046, which also makes the temporary 25 price difficult to reconcile with the adjacent observations.

This is an ex-post validation rule. In a live process I would quarantine the observation, compare it with another evaluated-price source, and release it only after confirmation. For the historical assignment, excluding the demonstrably isolated mark is preferable to allowing a known data error to determine the fitted curve and portfolio PnL.

2. Data and Sample Construction

The sample contains six large non-financial issuers with enough outstanding bonds to cover a useful duration range. I remove defaulted and convertible observations, require at least one year to maturity and roughly \$200 million outstanding, and keep the modal security level and rating category within each issuer-month. The isolated-mark screen is applied after those eligibility filters and before any model or return calculation.

Issuer	Obs.	Bonds	Median dur.	Duration range	Median spread
AAPL	1,474	52	7.31	0.95 - 20.99	19.0 bps
AMZN	908	33	6.58	0.95 - 20.03	15.0 bps
BA	1,304	51	7.39	0.92 - 16.41	28.5 bps
CAT	957	51	3.45	0.95 - 18.65	13.0 bps
DIS	1,237	44	8.82	0.96 - 18.19	20.0 bps
T	1,678	56	10.63	0.96 - 17.84	28.0 bps

Return treatment

Positions are formed at month-end and evaluated with the next observation for the same bond when the gap is 25 to 45 days. WRDS **RET_EOM** is the preferred monthly total return. If it is missing, the fallback combines clean-price return with coupon accrual using actual gap days over 360, consistent with the daily PnL convention Professor Boroditsky discussed in class.

The cleaning leaves 7,558 eligible bond-months and 7,224 valid next-month returns. The excluded Apple row is not replaced or interpolated.

Limits of the supplied fields

The files do not contain complete call schedules, bid-offer spreads, 144A status, subsidiary mapping, or borrow availability. I leave those as limitations rather than manufacture proxies that could create a second layer of measurement error.

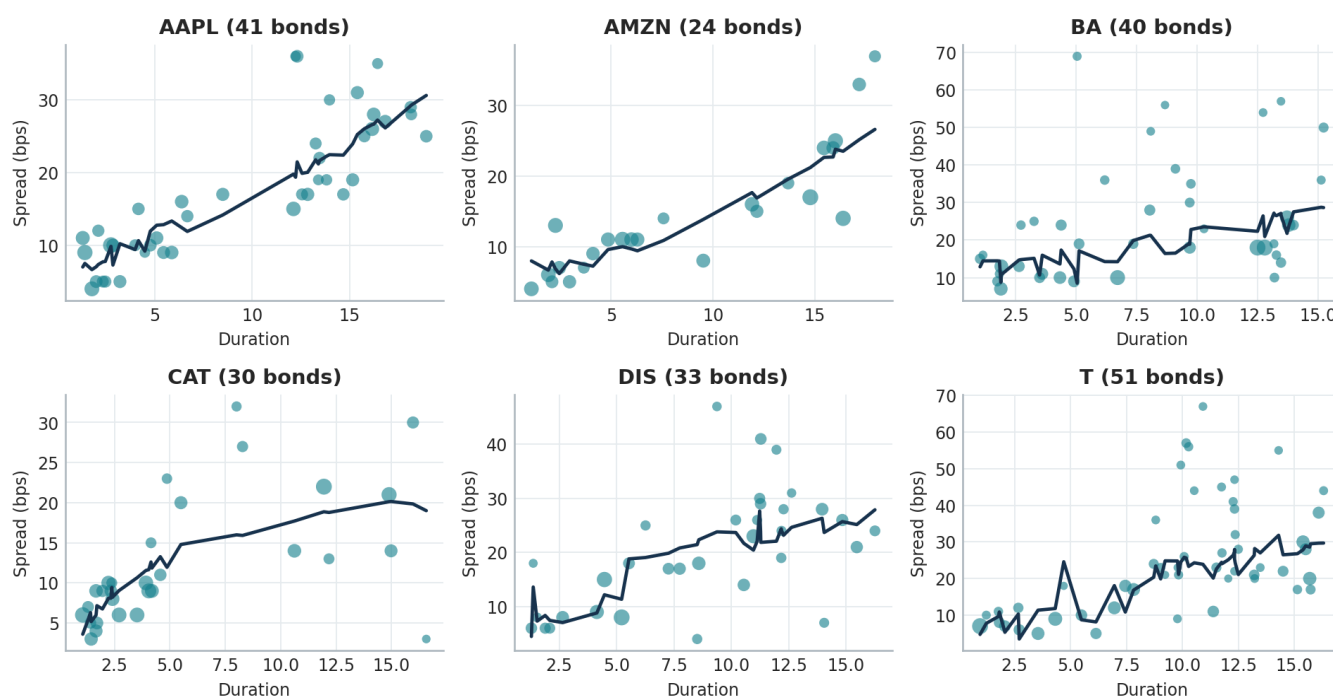
3. Issuer Spread Curves

For each issuer-month, I fit a transparent three-factor cross-sectional model:

$$T_Spread = b_0 + b_1 \log(1 + Duration) + b_2 Duration + b_3 Coupon$$

The regression weights combine TRACE dollar volume and amount outstanding. Professor Boroditsky emphasized that larger, actively traded issues are generally more reliable calibration points, so both inputs are winsorized within the issuer-month before weighting. A positive residual is wider than the fitted curve and classified as cheap; a negative residual is classified as rich.

Cleaned issuer spread curves on March 31, 2025



The latest common cross-section is March 31, 2025. The fit is strongest for AAPL and AMZN by median weighted R-squared, while BA and T show greater cross-sectional dispersion. This difference is one reason to rank residuals within issuer rather than compare raw spread gaps across issuers.

4. Portfolio Construction

Each issuer is treated as its own relative-value sleeve. The strategy buys bonds with positive residuals, shorts bonds with negative residuals, and sizes the short side so the signed market-value duration contribution is approximately zero. Each issuer sleeve is normalized to one dollar of gross market value, and the available sleeves are equal-weighted each month.

$$\text{sum}(\text{Position} \times \text{Percentage Price} \times \text{Duration}) = 0$$

Three passes

Pass	Selection rule	Purpose
1. Baseline top 3	Three cheapest and three richest; size by residual strength	Establish the assignment's direct rich-cheap rule
2. Concentrated top 2	Two strongest bonds on each side	Test whether weaker selected signals dilute the trade
3. Liquidity-aware top 3	Three per side; residual score adjusted by TRACE dollar volume	Prefer signals that may be easier to implement

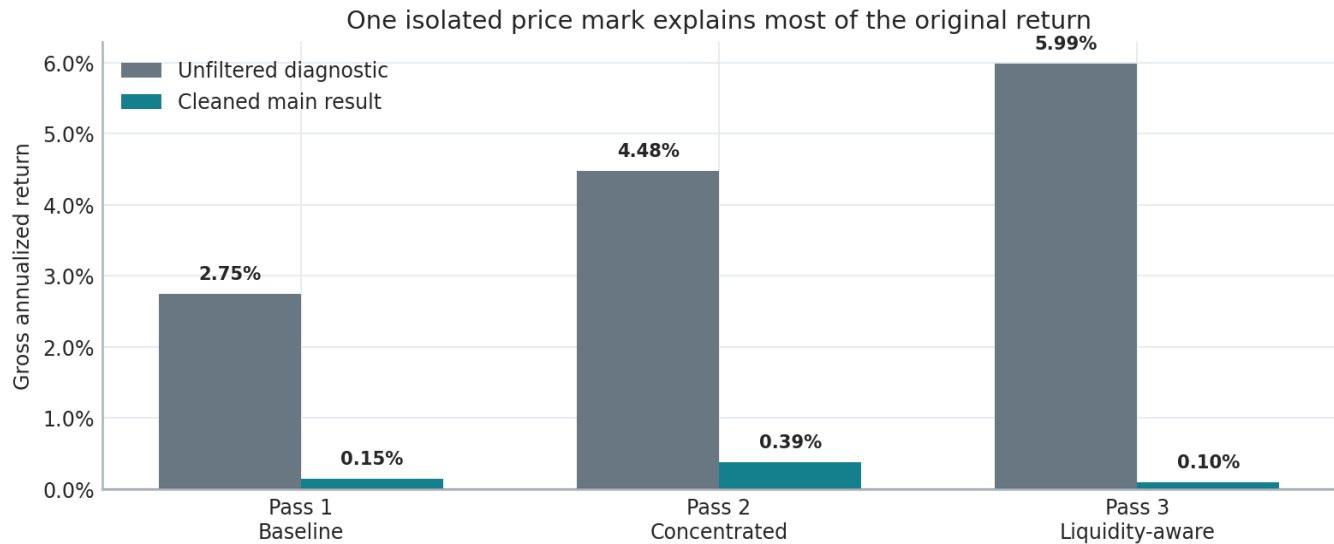
What duration neutrality does and does not do

The hedge removes first-order exposure to a parallel interest-rate move. It does not neutralize Treasury-curve slope or curvature, spread duration, issuer beta, downgrade risk, liquidity exposure, financing, or short-borrow constraints. Those remaining risks matter because the expected cleaned return is small.

Model selection

I compare all three rules on the same sample and select using the 25 bp cost-adjusted Sharpe ratio. This is useful for the assignment's three-pass exercise, but it introduces selection bias. A live research process should freeze the rule on an earlier training period and evaluate it on untouched later data.

5. What the Data Correction Changes



Portfolio pass	Raw gross	Clean gross	Raw net 25 bp	Clean net 25 bp
1. Baseline top 3	2.75%	0.15%	0.45%	-2.10%
2. Concentrated top 2	4.48%	0.39%	2.02%	-1.98%
3. Liquidity-aware top 3	5.99%	0.10%	3.60%	-2.17%

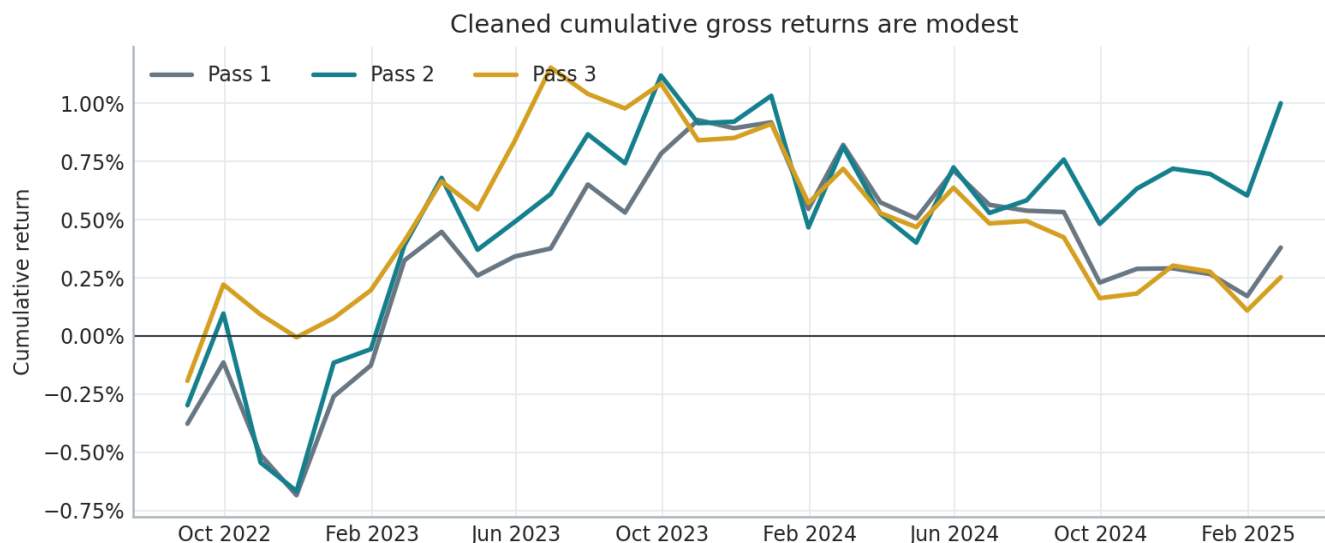
Interpretation

The raw backtest says that every design improves, culminating in a 5.99% gross annualized return for the liquidity-aware pass. Once the isolated Apple mark is removed, that ranking does not hold: Pass 2 leads at only 0.39%, while Pass 3 falls to 0.10%. The original improvement was therefore not evidence that liquidity weighting produced a stronger signal. It was mainly an interaction between the selected Apple bond and a bad month-end price.

I keep the raw results in the notebook and this table for transparency, but I do not use them as performance evidence.

6. Cleaned Portfolio Performance

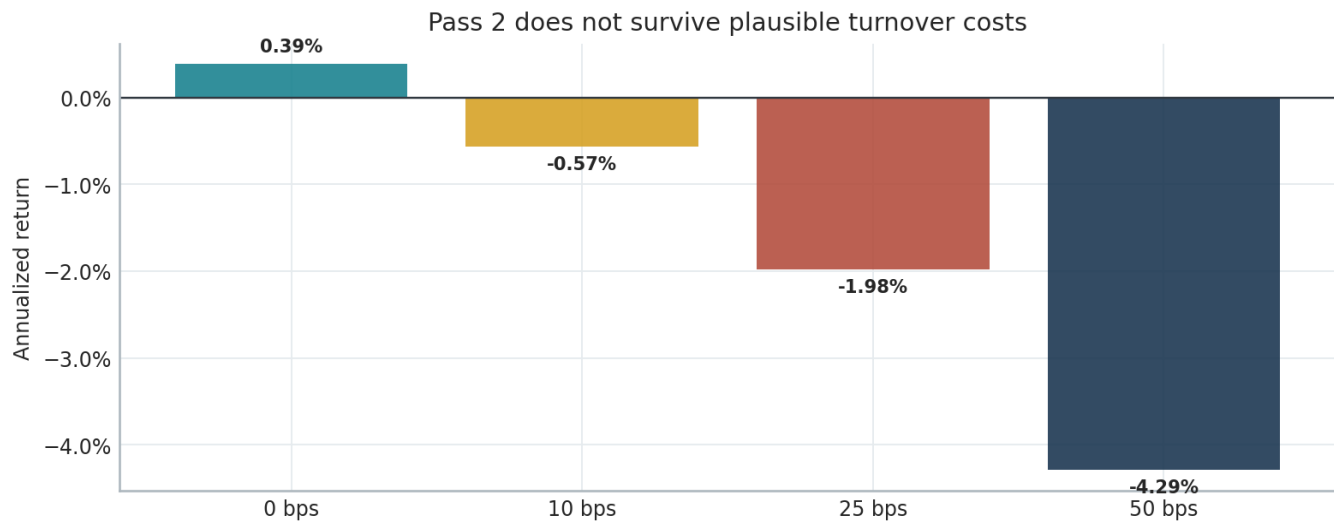
Pass 2 has the least-negative 25 bp cost-adjusted Sharpe ratio and is therefore the selected clean specification. Its gross result is modest: 0.39% annualized return, 1.02% annualized volatility, and a 0.38 Sharpe ratio over 31 monthly observations.



Issuer	Ann. return	Ann. vol.	Sharpe	Total return	Win rate	Turnover
AAPL	0.39%	1.11%	0.35	1.01%	51.6%	0.87
AMZN	-0.39%	0.79%	-0.50	-1.02%	51.6%	0.79
BA	0.73%	1.90%	0.39	1.90%	48.4%	0.79
CAT	-0.60%	3.66%	-0.16	-1.55%	58.1%	0.79
DIS	1.66%	1.59%	1.05	4.36%	67.7%	0.86
T	0.42%	3.16%	0.13	1.08%	45.2%	0.82

The combined best month is December 2022 at 0.55%; the worst is October 2022 at -0.64%. Excluding AAPL now leaves annualized return at 0.38%, close to the all-issuer result. That is a useful cross-check: the cleaned performance is no longer driven by one issuer.

7. Trading Costs and Residual Characteristics



Average one-way monthly turnover is 0.82. Even a 10 bp assumption turns annualized return negative; at 25 bp it is -1.98%. The hedge is mechanically tight, with average absolute net DV01 effectively zero on the \$1 million gross-capital convention, but good hedging cannot rescue a signal whose gross return is smaller than implementation cost.

What is special about selected bonds?

Group	Obs.	Median abs. residual	Median dollar volume	Median amount out.	Median dur.	Median TMT
Long Cheap	372	27.2 bps	\$17,578,816	\$500,000	12.68	20.23
Not Selected	6,814	4.2 bps	\$50,222,070	\$1,100,000	7.11	8.71
Short Rich	372	15.5 bps	\$21,082,954	\$758,680	13.66	24.25

The long-cheap bonds are less liquid, smaller, and longer-dated than the non-selected observations. This supports a cautious reading of the residual: part of the apparent cheapness may compensate investors for liquidity and long-horizon risks rather than represent a temporary pricing mistake.

8. Conclusion

The assignment's main lesson changed once I investigated the early-2025 jump. The Apple observation is not merely a large but plausible return; it is a temporary month-end price collapse surrounded by stable prices. Because it distorts both the cross-sectional curve and two adjacent return intervals, the correct sequence is to quarantine it first, then refit the curve and rebuild returns.

After cleaning, all three portfolio designs have small positive gross returns and negative returns after a 25 bp turnover charge. I would not present the strategy as profitable in this sample.

What I would do next

- Validate month-end prices against a second source before they enter research data.
- Define the quality-control thresholds in advance and log every exclusion.
- Freeze the portfolio rule on a training period, then evaluate it out of sample.
- Add bid-offer estimates, financing, borrow availability, call schedules, and issuer-entity mapping.
- Test key-rate and spread-duration hedges in addition to parallel-rate DV01 neutrality.

Assumptions and limitations

The isolated-mark rule uses the next monthly observation and is therefore an ex-post historical validation check. The curve is a transparent regression rather than a cash-flow model. Return timing assumes the remaining month-end fields were available at formation. Trading costs are stylized, and actual shorting constraints may be materially worse.

References

- Boroditsky, M. MTH9897 Quantitative Investment Framework, class notes for sessions 1-2, 2026.
- Nelson, C. R., and Siegel, A. F. (1987). Parsimonious Modeling of Yield Curves. *Journal of Business*, 60(4), 473-489.
- Wharton Research Data Services. WRDS Bond Returns documentation:
wrds-www.wharton.upenn.edu/pages/get-data/wrds-bond-returns/wrds-bond-returns/
- Apple Inc. pricing term sheet for the 4.65% note due 2046, CUSIP 037833 BX7: [SEC filing](#).